Oxford College of Emory University

Chapter 15: Monetary Policy

Part 2: Policy in Practice

ECON_OX 112 · Principles of Macroeconomics

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Roadmap

- Choose a policy rate.
- Explain the lower bound and QE.
- Compare short- and long-run effects.

Inflation and the Output Gap

Inflation target

A goal for inflation over time.

Output gap

Real GDP relative to potential GDP.

$g = 100 \times (Y - Y^*) / Y^*$

Y: real GDP · Y*: potential GDP · g: output gap (%)

$Y = 980$; $Y^* = 1{,}000 \rightarrow g = -2\%$

Fed inflation goal: 2% PCE.

Federal Reserve Board: inflation objective · Federal Reserve Board: policy rules

The Taylor Rule

$$i = r^* + \pi + 0.5(\pi - \pi^*) + 0.5g$$

r*: neutral real rate	π: inflation	π*: target	g: output gap
2%	4%	2%	−2%

Inflation above target: Raises the suggested rate.

Output below potential: Lowers the suggested rate.

Calculate i. Compare your reasoning with a partner.

$i = 2 + 4 + 1 - 1 = 6\%$

i is a nominal policy-rate benchmark.

Federal Reserve Board: policy rules

The Zero Lower Bound

Limit

Nominal rates cannot fall far below zero.

Reason

People can hold cash instead.

Next tool

QE targets longer-term borrowing costs.

Some central banks have used slightly negative rates.

Federal Reserve Board: policy limits

Quantitative Easing

Action

The central bank buys large amounts of longer-term bonds.

Bond market

Higher bond prices mean lower yields.

Spending

Lower borrowing costs can support C and I.

More reserves do not automatically mean more lending.

Federal Reserve Board: asset purchases and longer-term rates

Short- and Long-Run Effects

	Short run	Long run
Wages and prices	Sticky	Fully adjusted
Real output	Can change	At potential
Price level	Starts adjusting	Reflects money supply

One-time money change; productive capacity fixed.

ECB: short-run effects and long-run neutrality

Monetary Neutrality

Long run

Money changes prices, not real output.

Real capacity

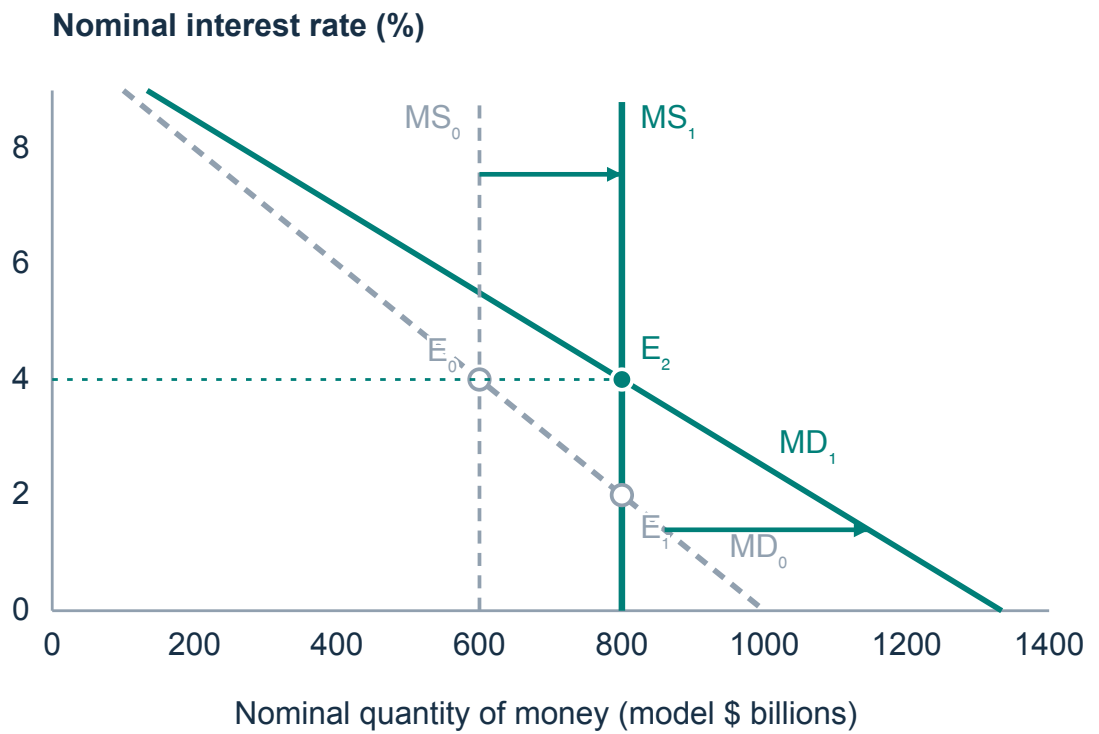
Workers, capital and technology determine potential output.

Money supply	Price level	Real output
+10%	+10%	Unchanged

Full price adjustment; other long-run conditions fixed.

ECB: short-run effects and long-run neutrality

Monetary Expansion



Start

MS = 600; i = 4%.

Short run

MS rises to 800; i falls to 2%.

Long run

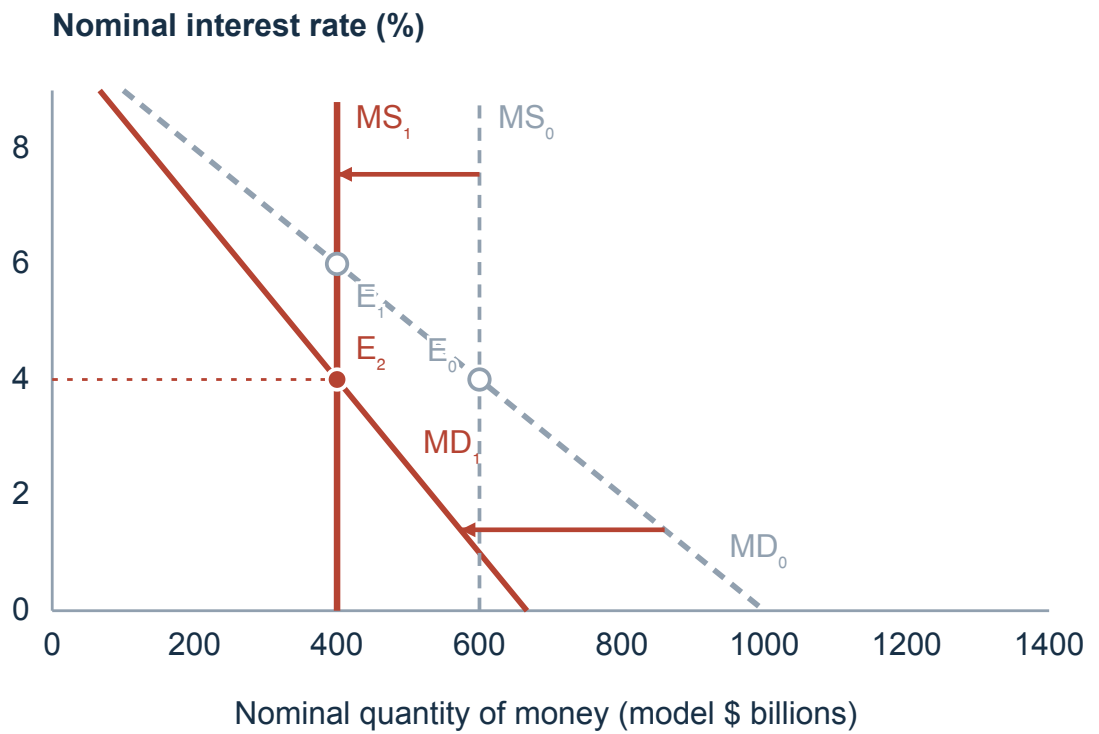
Prices rise. MD shifts right; i returns to 4%.

One-time MS change; long-run expected inflation unchanged.

New MS stays at 800; potential output is unchanged.

ECB: short-run effects and long-run neutrality

Monetary Contraction



Start

MS = 600; i = 4%.

Short run

MS falls to 400; i rises to 6%.

Long run

Prices fall. MD shifts left; i returns to 4%.

One-time MS change; long-run expected inflation unchanged.

New MS stays at 400; potential output is unchanged.

ECB: short-run effects and long-run neutrality

Long-Run Checkpoint

MS falls 20%. Prices fully adjust; other long-run conditions stay fixed.

Predict output, prices, and MD. Which curve moves first?

Real output	Price level	Nominal MD
Unchanged	−20%	Shifts left

MS moves first. Lower prices later shift MD left.

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